PHYSICAL MODELS & ERROR ANALYSIS

Dynamical Modelling and Validation of Long-Term LEO Trajectory Prediction

The force models, their governing equations, and the measured accuracy against Sentinel-3 precise orbits

Project: URSC — modular LEO orbit propagation & determination (Java, zero runtime dependencies)

Validation data: Sentinel-3A / Sentinel-3B POEORB precise orbits (GPS-derived POD), 16 Oct 2024, 10 s sampling

Document type: Physics & error-analysis report

Reference architecture: Orekit (independent clean-room implementation)

Abstract. This report documents the physics of a from-scratch orbital dynamics engine for low-Earth-orbit (LEO) trajectory prediction and the accuracy it achieves against independent, GPS-derived precise orbits. The equations of motion integrate a full perturbation stack: a spherical-harmonic geopotential (EGM96 to degree/order 70), luni-solar third-body attraction, atmospheric drag through the NRLMSISE-00 thermosphere, solar radiation pressure with a conical eclipse model, solid and ocean tides (IERS-2010 / FES2004), and post-Newtonian relativistic corrections, all referred to a rigorous GCRF ↔ ITRF transformation (IAU-2006/2000B precession, nutation, Earth rotation and polar motion). Validated against Sentinel-3 POEORB ephemerides, pure propagation from a single state vector reproduces the precise orbit with radial and cross-track errors held to a few metres over 24 h, while the along-track channel — the integrated signature of residual non-conservative mismodelling — grows to roughly 0.3 km. A batch least-squares fit reduces the 6-hour residual from ≈40 m to ≈5 m RMS. The results confirm that the conservative field is modelled correctly and localise the remaining error physically in the drag/ballistic channel.

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1 Physical setting: the LEO perturbation environment

A satellite in low Earth orbit does not move on a closed Keplerian ellipse. The dominant central attraction is perturbed by a hierarchy of smaller forces whose relative importance is strongly altitude-dependent. The two validation targets, Sentinel-3A and Sentinel-3B, fly in a near-polar, Sun-synchronous orbit at roughly 800 km altitude (inclination ≈98.6°, near-circular), representative of Earth-observation LEO.

The purpose of the engine is twofold: (i) **prediction** — integrate the state forward over hours to years for trajectory and orbital-lifetime studies; and (ii) **determination** — recover the initial state and dynamical parameters that best reproduce a set of observations. Both rest on the same force models, so the physical fidelity of those models sets the ceiling on achievable accuracy.

Table 1. Orbital characteristics of the validation targets (from the POEORB seed state).

Satellite	Perigee altitude	Inclination	Eccentricity	Period	POEORB sampling
Sentinel-3A	780.3 km	98.618°	0.00212	~100.9 min	10 s, 26 h arc
Sentinel-3B	786.9 km	98.625°	0.00065	~101.0 min	10 s, 26 h arc

> **Why POEORB is a strong reference.** The Sentinel-3 POEORB product is a *precise* orbit determined on the ground from the satellite's onboard GPS receiver (dual-frequency POD), with a quoted accuracy of a few centimetres (radial < 3 cm). It is therefore effectively "truth" at the metre level relevant here, and — crucially — it is *independent* of our dynamics: it is a GNSS measurement product, not another propagation. This makes it an honest external check of the physics.

2 Equations of motion and the perturbation hierarchy

The satellite's motion is integrated in the geocentric celestial reference frame (GCRF), a pseudo-inertial frame, as a second-order system for the position r and velocity v. The total acceleration is the two-body term plus the sum of perturbations:

$$\ddot{\vec{r}} = -\frac{\mu}{r^3}\vec{r} + \vec{a}_{\mathrm{NS}} + \vec{a}_{3B} + \vec{a}_{\mathrm{drag}} + \vec{a}_{\mathrm{SRP}} + \vec{a}_{\mathrm{tide}} + \vec{a}_{\mathrm{rel}} \quad (1)$$

where the perturbing accelerations are, in turn, the non-spherical geopotential (a_{NS}), third-body attraction (a_{3B}), atmospheric drag (a_{drag}), solar radiation pressure (a_{SRP}), solid and ocean tides (a_{tide}), and relativistic corrections (a_{rel}). Each is an independent, individually switchable physical model. Table 2 orders them by characteristic magnitude at the Sentinel-3 altitude; this hierarchy is what the validation later confirms.

Table 2. Order-of-magnitude of each acceleration at ~800 km, with the dominant orbital effect.

Perturbation	Magnitude (m s^{-2})	Relative to central	Dominant effect on the orbit
Central attraction $-\mu r/r^3$	~8.2	1	Keplerian ellipse
J_2 (oblateness)	~1×10^{-2}	10^{-3}	Nodal regression, apsidal rotation
Higher geopotential ($n\geq3$)	~1×10^{-5}	10^{-6}	Short-period + resonant terms
Luni-solar third body	~5×10^{-6}	10^{-6}	Secular/long-period on Ω, *i*, *e*
Atmospheric drag	~1×10^{-7}	10^{-8}	**Secular along-track decay**
Solar radiation pressure	~1×10^{-7}	10^{-8}	Periodic; eclipse-modulated
Solid Earth tides	~2×10^{-9}	10^{-10}	Small periodic geopotential change
Ocean tides	~1×10^{-9}	10^{-10}	Small periodic geopotential change
Relativity (Schwarzschild)	~3×10^{-10}	10^{-11}	Apsidal precession (~few m/day)
Lense–Thirring (frame drag)	~1×10^{-12}	10^{-13}	Nodal drag (negligible for LEO)

The central and J_2 terms govern the gross geometry; everything below drag is conservative and cleanly periodic, which is why — as the validation shows — it maps into bounded radial and cross-track errors. Drag is the only large *dissipative* term, and it is the hardest to model because it depends on the state of the thermosphere.

3 The gravitational field

The Earth's gravity is derived from a potential expanded in spherical harmonics. In body-fixed spherical coordinates (radius r, geocentric latitude φ, longitude λ):

$$U(r,\phi,\lambda) = \frac{\mu}{r}\left[1 + \sum_{n=2}^{N}\sum_{m=0}^{n}\left(\frac{R_\oplus}{r}\right)^n \bar{P}_{nm}(\sin\phi)\left(\bar{C}_{nm}\cos m\lambda + \bar{S}_{nm}\sin m\lambda\right)\right] \quad (2)$$

Here $R_\oplus$ is the field's reference radius, $\bar{P}_{nm}$ are the fully-normalised associated Legendre functions, and $\bar{C}_{nm}$, $\bar{S}_{nm}$ are the normalised Stokes coefficients. The acceleration is ∇U. The zonal term n=2, m=0 is the oblateness ($J_2 = -C_{20}\sqrt{5} \approx 1.0826\times10^{-3}$), which produces the Sun-synchronous nodal regression that defines this orbit class.

3.1 Coefficient set and normalisation

The field is the **EGM96** satellite-only solution, complete to degree and order 70, with GM = 3.986004415×10^{14} m^{3}s^{-2} and $R_\oplus$ = 6 378 136.3 m. The published coefficients are fully normalised; they are converted to the unnormalised form used in the recursion by

$$\{\bar{C}_{nm}, \bar{S}_{nm}\} = \left[\frac{(n+m)!}{(2-\delta_{0m})(2n+1)(n-m)!}\right]^{1/2} \{C_{nm}, S_{nm}\} \tag{3}$$

with δ_{0m} the Kronecker delta. Default runs truncate at degree/order 30, which captures essentially all of the geopotential signal relevant over multi-day LEO arcs; the full 70×70 field is available when required.

3.2 Evaluation method

The potential gradient is computed with the **Cunningham–Gottlieb** recursion for the normalised potential derivatives (Montenbruck & Gill, *Satellite Orbits*, §3.2.4), which is numerically stable to high degree and avoids the explicit evaluation of Legendre polynomials and their derivatives. The field is static only in the Earth-fixed frame, so the acceleration is evaluated in ITRF and rotated back to GCRF using the frame transformation of §9. The expansion may be truncated independently in degree and order, allowing, for example, a zonal-only J_2–J_6 field (order 0) for secular studies or a full tesseral field for high-fidelity work.

4 Third-body (luni-solar) attraction

The Sun and Moon perturb the geocentric orbit both by attracting the satellite directly and by attracting the Earth itself. Writing the satellite position r and the third body's geocentric position s, the perturbing acceleration is the difference of these two terms:

$$\vec{a}_{3B} = GM_b\left(\frac{\vec{s}-\vec{r}}{|\vec{s}-\vec{r}|^3} - \frac{\vec{s}}{|\vec{s}|^3}\right) \tag{4}$$

The bracketed difference is the standard formulation: the first term is the direct attraction on the satellite, the second (the "indirect" term) removes the acceleration of the frame origin so that the result is the *tidal* acceleration relative to the Earth. Gravitational constants used are $GM_\odot = 1.32712440018\times10^{20}$ and $GM_\leftmoon = 4.9028\times10^{12}$ m³s^{-2}. The solar and lunar positions come from compact analytical ephemeris series (Montenbruck & Gill), accurate to ≈0.1° / a few hundred kilometres — far better than the 10^{-6} relative precision needed for the perturbation itself.

5 Atmospheric drag and the thermosphere

Drag is the dominant non-conservative force at this altitude and, as the validation confirms, the single largest contributor to prediction error. The acceleration opposes the velocity of the satellite relative to the co-rotating atmosphere:

$$\vec{a}_{\text{drag}} = -\frac{1}{2}\frac{C_D A}{m}\,\rho\,|\vec{v}_{\text{rel}}|\,\vec{v}_{\text{rel}}, \qquad \vec{v}_{\text{rel}} = \vec{v} - \vec{\omega}_\oplus \times \vec{r} \tag{5}$$

where ρ is the local mass density, C_D the drag coefficient, A the cross-sectional area, m the mass, and $\boldsymbol{\omega}_\oplus = 7.292115\times10^{-5}$ rad s^{-1} the Earth rotation rate. The grouping C_DA/m is the **ballistic coefficient**; it, together with the density ρ, carries essentially all of the modelling uncertainty — the geometry of Eq. (5) is exact.

5.1 Density model: NRLMSISE-00

Density is supplied by a pluggable atmosphere interface; the high-fidelity option is the **NRLMSISE-00** empirical thermosphere model (a clean-room port of the reference C implementation, verified against the model's official test case to seven significant figures). It returns the effective total mass density for satellite drag (including anomalous oxygen above ~500 km) as a function of altitude, latitude, longitude, day-of-year, local solar time, and the solar/geomagnetic activity drivers:

- **F10.7** — the daily 10.7 cm solar radio flux (a proxy for EUV heating);
- **F10.7a** — its 81-day centred average;
- **Ap** — the daily planetary geomagnetic index.

These drivers enter through a *space-weather provider* that is either a constant (the validation used F10.7 = 220, F10.7a = 205, Ap = 12, representative of the active 2024 solar maximum) or a file-backed historical series (CelesTrak CSSI format). Lower-fidelity alternatives — a static exponential atmosphere and the diurnal Harris–Priester model — are also available for comparison and fast studies.

> **Physical caveat.** Thermospheric density is uncertain at the 10–30% level even with a perfect activity input, because NRLMSISE-00 is a climatological average. Combined with the use of *constant* rather than true daily space-weather indices and a fixed cross-sectional area (the real satellite's attitude changes A), this is the principal reason the along-track error grows as it does in §10.

6 Solar radiation pressure and eclipse geometry

Sunlight carries momentum; its absorption and reflection push the satellite away from the Sun:

$$\vec{a}_{\mathrm{SRP}} = -\,\nu\, C_R\, \frac{A}{m}\, P_0 \left(\frac{\mathrm{AU}}{d}\right)^2 \hat{\vec{u}}_{\odot \to \mathrm{sat}} \tag{6}$$

where $P_0 = 4.56\times10^{-6}$ N m^{-2} is the solar radiation pressure at 1 AU, d the Sun–satellite distance, C_R the reflectivity coefficient (1 = fully absorbing, 2 = fully reflecting; the validation used 1.3), and $\nu \in [0,1]$ the fraction of the solar disc visible from the spacecraft. The $(\mathrm{AU}/d)^2$ factor is the inverse-square fall-off of the flux.

6.1 Conical shadow model

The shadow factor ν is computed from a **conical** umbra/penumbra model using the apparent angular radii of the Sun and the occulting Earth as seen from the spacecraft, $\alpha_\odot = \arcsin(R_\odot/d)$ and $\alpha_\oplus = \arcsin(R_\oplus/r)$, and their angular separation. Full sunlight ($\nu = 1$), total umbra ($\nu = 0$), and the partial penumbra (computed from the lens-shaped overlap area of the two apparent discs, with the annular case handled separately) are all distinguished. This smooth shadow function is important physically: SRP switches on and off twice per orbit, and a sharp (cylindrical) shadow would inject spurious along-track impulses at each crossing.

7 Solid Earth and ocean tides

The Sun and Moon deform the solid Earth and redistribute ocean mass; both change the geopotential coefficients slightly, which in turn perturbs the orbit.

7.1 Solid Earth tides

Implemented to the IERS-2010 Conventions, frequency-independent (Step 1) model: the tide-raising bodies induce corrections to the normalised coefficients for degrees 2 and 3, plus the degree-4 response to the degree-2 tide (the k^+ terms):

$$\Delta\bar{C}_{nm} - i\,\Delta\bar{S}_{nm} = \frac{k_{nm}}{2n+1}\sum_b \frac{GM_b}{GM_\oplus}\left(\frac{R_\oplus}{r_b}\right)^{n+1} \bar{P}_{nm}(\sin\phi_b)\, e^{-im\lambda_b} \tag{7}$$

with nominal (real) Love numbers from IERS-2010 Table 6.3 (k_{20} = 0.29525, k_{21} = 0.29470, k_{22} = 0.29801, etc.). The sum runs over the Moon and Sun; r_b, φ_b, λ_b are the body's body-fixed radius, latitude and longitude. The small frequency-dependent (Step 2) and anelastic (imaginary) terms are omitted as negligible for LEO trajectory work.

7.2 Ocean tides

Implemented to IERS-2010 Eq. 6.15, driven by a bundled truncation of the **FES2004** model — the eight dominant constituents (Q_1, O_1, P_1, K_1, N_2, M_2, S_2, K_2) to degree and order 4:

$$\Delta\bar{C}_{nm} = \sum_f \left[(C^+_{f,nm} + C^-_{f,nm})\cos\theta_f + (S^+_{f,nm} + S^-_{f,nm})\sin\theta_f \right] \tag{8}$$

where θ_f is the Doodson argument of constituent f, assembled from the fundamental (Delaunay/Doodson) astronomical arguments. Both tidal contributions are at the 10^{-9} m s^{-2} level — too small to see in the current metre-level validation, but included for completeness and precise long-arc work.

8 Relativistic corrections

General relativity modifies the central attraction. The dominant (Schwarzschild) correction, from the IERS-2010 Conventions (Eq. 10.12) with PPN parameters $\beta = \gamma = 1$, is always included:

$$\vec{a}_{\text{Schw}} = \frac{\mu}{c^2 r^3}\left[\left(\frac{4\mu}{r} - v^2\right)\vec{r} + 4(\vec{r}\cdot\vec{v})\,\vec{v}\right] \tag{9}$$

It is of order $\mu v^2/(c^2 r^2) \sim 3\times10^{-10}$ m s^{-2} and acts chiefly to precess the apsides (a few metres per day). An optional Lense–Thirring (frame-dragging) term from the Earth's rotation is available,

$$\vec{a}_{\text{LT}} = \frac{2\mu}{c^2 r^3}\left[\frac{3}{r^2}(\vec{r}\cdot\vec{J})(\vec{r}\times\vec{v}) + (\vec{v}\times\vec{J})\right] \tag{10}$$

where **J** is the Earth's angular momentum per unit mass ($\approx 9.8\times10^8$ m^{2}s^{-1}). It is $\sim10^{-12}$ m s^{-2} at LEO — negligible for prediction but retained for completeness. The de Sitter (geodesic precession) term, which requires the Earth's heliocentric velocity, is omitted as sub-mm s^{-2}.

9 Reference frames and time scales — the GNSS connection

Correctly relating the inertial integration frame (GCRF) to the Earth-fixed frame (ITRF) is essential: the geopotential and atmosphere are naturally expressed in ITRF, and — directly relevant to GNSS — the Sentinel-3 POEORB state

vectors are *given* in ITRF. ITRF is the same realisation family in which GPS/GNSS products are expressed, so the frame chain below is exactly the transformation a GNSS positioning pipeline must also apply.

9.1 GCRF ↔ ITRF transformation

The full rotation follows the IAU-2006/2000B conventions as a chain of bias, precession, nutation, Earth rotation and polar motion:

$$\vec{r}_{\mathrm{ITRF}} \;=\; W(x_p, y_p)\, R_3(\mathrm{GAST})\, N(\Delta\psi, \Delta\varepsilon)\, P(t)\, B\, \vec{r}_{\mathrm{GCRF}} \tag{11}$$

- **B** — frame bias (GCRF to the mean J2000 dynamical frame);
- **P(t)** — precession via the IAU-2006 Fukushima–Williams angles;
- **N(Δψ,Δε)** — nutation via the IAU-2000B 77-term luni-solar series (accuracy ~1 mas), with mean obliquity ε_A;
- **R_3(GAST)** — diurnal Earth rotation through the Greenwich Apparent Sidereal Time (mean sidereal time plus the equation of the equinoxes);
- **W(x_p,y_p)** — polar motion (a small-angle wobble matrix).

The implementation follows the SOFA reference algorithm and reproduces SOFA to ~10^{-13} radian in the unit tests. A simpler sidereal-only model (GMST rotation, IAU-82 polynomial, Eq. 12) is also provided for fast runs where sub-arcsecond fidelity is not required:

$$\theta_{\mathrm{GMST}} = 280.46061837^\circ + 360.98564736629^\circ\, d + 3.8793 \times 10^{-4}\, T^2 - \frac{T^3}{3.871 \times 10^7} \tag{12}$$

9.2 Earth orientation and time scales

Earth-orientation parameters (EOP: UT1−UTC, polar motion x_p, y_p) enter through a provider. For the Sentinel-3 validation the UT1−UTC offset carried in the POEORB file itself (ΔUT1 = 0.0579 s) was supplied, which fixes the Earth-rotation phase correctly; polar motion was set to zero. The time scales are related by

$$\mathrm{TT} = \mathrm{TAI} + 32.184\ \mathrm{s}, \qquad \mathrm{UTC} = \mathrm{TAI} - \Delta AT, \qquad \mathrm{UT1} = \mathrm{UTC} + \Delta UT1 \tag{13}$$

with ΔAT the integer count of leap seconds (full 1972–2017 table embedded) and ΔUT1 the sub-second UT1 offset. Terrestrial Time (TT = TAI + 32.184 s) is the argument of the precession/nutation series; UT1 drives Earth rotation; UTC is the civil clock in which the observations are tagged. Dates are stored as continuous, leap-second-free seconds past the J2000 epoch, so time differences are exact across leap-second boundaries.

10 Validation against Sentinel-3 precise orbits

The validation methodology is deliberately conservative and transparent. The propagator is **seeded from a single POEORB state vector** (the first epoch), transformed ITRF → GCRF with the file's own UT1, and integrated forward under the full force model with *no* fitting. At each subsequent epoch the propagated position is differenced against the precise orbit, and the error vector is decomposed into the local **radial–along–cross (RTN)** orbital frame:

$$\hat{\vec{R}} = \frac{\vec{r}}{|\vec{r}|}, \qquad \hat{\vec{C}} = \frac{\vec{r} \times \vec{v}}{|\vec{r} \times \vec{v}|}, \qquad \hat{\vec{A}} = \hat{\vec{C}} \times \hat{\vec{R}} \tag{14}$$

This decomposition is physically diagnostic: different force errors project differently onto these axes. Energy/period (drag) errors accumulate *along-track*; mismodelled oblateness or out-of-plane forces appear *cross-track*; and a radial bias indicates a semi-major-axis or central-term error.

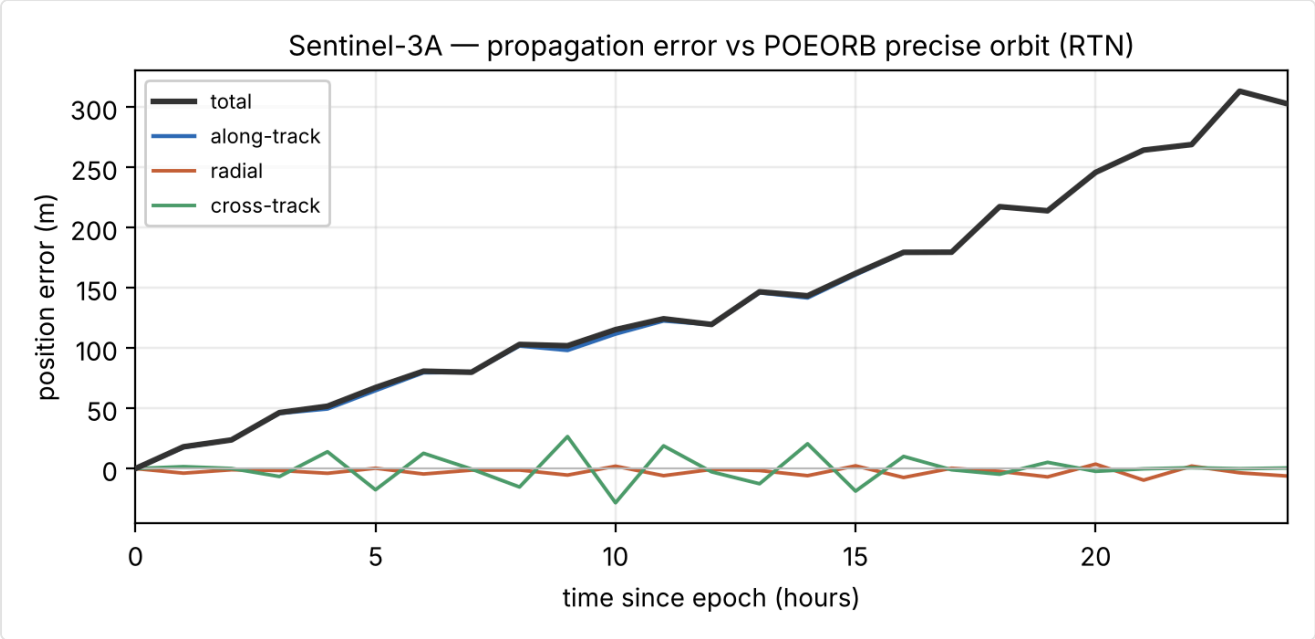


Figure 1. Sentinel-3A — RTN components of the propagation error vs the POEORB precise orbit over 24 h (full force model, no fit). Radial (orange) and cross-track (green) stay within ±30 m; the along-track channel (blue, hidden under the black total) dominates.

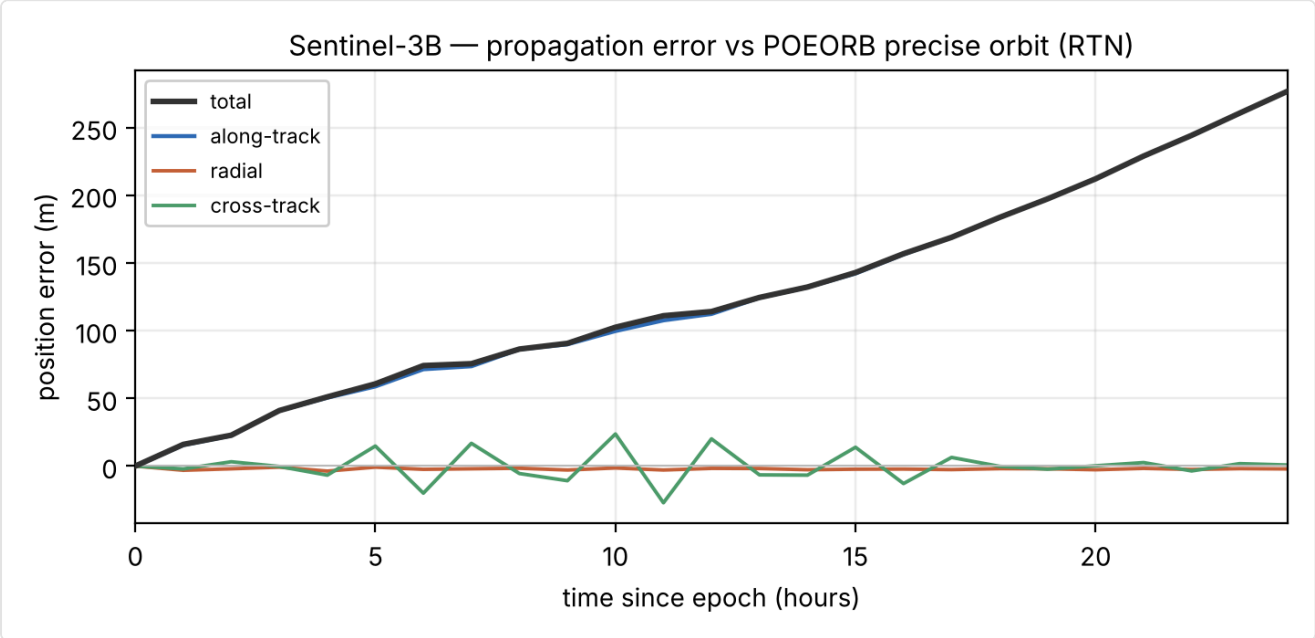


Figure 2. Sentinel-3B — the same analysis. The behaviour is quantitatively consistent with S3A: a few-metre conservative error floor and a monotonically growing along-track term.

Table 3. Measured error summary (pure propagation, full model, no estimation).

Satellite	Span	Radial (max)	Cross-track (max)	Along-track (max)	Total RMS	Total max
Sentinel-3A	12 h	~6 m	~28 m	~123 m	81.8 m	131 m
Sentinel-3A	24 h	~10 m	~28 m	~313 m	165.6 m	315 m
Sentinel-3B	24 h	~4 m	~27 m	~277 m	146.2 m	277 m

> **Physical reading of the result.** Radial and cross-track errors remain at the *metre* level over an entire day. These axes are driven by the conservative field — geopotential, third body, tides, relativity — and by the frame transformation. Holding them to a few metres is direct evidence that the gravity field, the luni-solar model, and the GCRF ↔ ITRF chain are implemented correctly. The error lives almost entirely **along-track**, exactly where a small dissipative (drag/ballistic) acceleration error must accumulate.

11 Orbit determination and drag-coefficient observability

To separate a genuine dynamical error from a merely imperfect initial state, a **batch least-squares** fit was run: the initial state (and optionally the drag coefficient) is adjusted to minimise the position residual against the POEORB arc. Treating each precise-orbit epoch as a position observation ρ_i, the fit minimises the RMS

$$\mathrm{RMS} = \left[\frac{1}{M} \sum_{i=1}^{M} \left| \vec{r}_i^{\,\mathrm{comp}} - \vec{r}_i^{\,\mathrm{obs}} \right|^2 \right]^{1/2} \tag{15}$$

via Gauss–Newton iteration on the (finite-difference, scaled) normal equations with small Levenberg–Marquardt damping (see the technical report for the numerics). The results over a 6-hour arc (position observations every 120 s):

Table 4. Batch least-squares residual reduction over a 6 h arc (gravity 20×20).

Satellite	Estimated	Pre-fit RMS	Post-fit RMS	Improvement	Est. C_D
Sentinel-3A	state (6)	38.4 m	4.8 m	8.0×	—
Sentinel-3A	state + C_D	38.4 m	4.5 m	8.6×	5.26 (!)
Sentinel-3B	state (6)	41.2 m	5.9 m	6.9×	—
Sentinel-3B	state + C_D	41.2 m	5.2 m	8.0×	−3.20 (!)

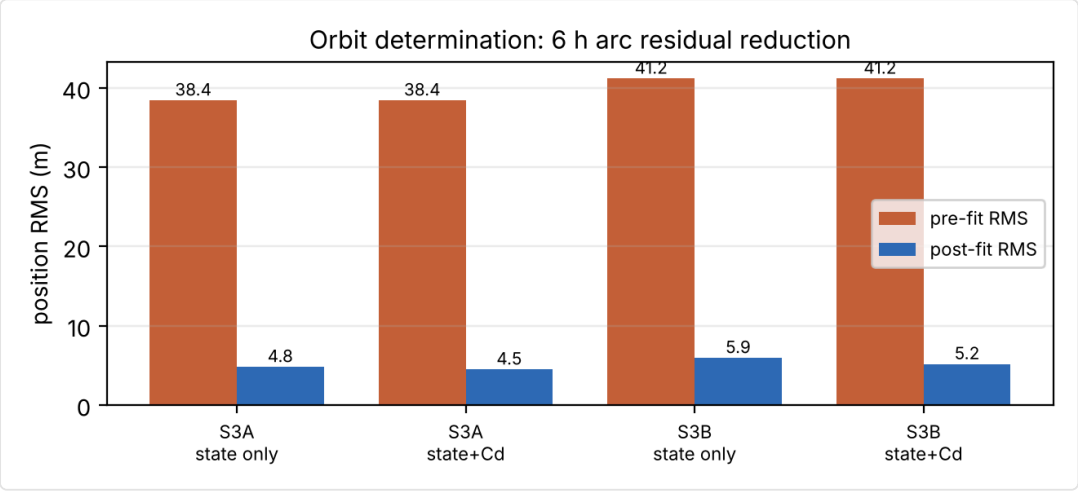


Figure 3. Residual RMS before and after the batch fit. Adjusting the initial state alone removes most of the 6-hour residual; adding C_D helps only marginally.

Honest finding — drag-coefficient observability. The state-only fit is robust (residual collapses by 7–8× to ~5 m). However, the estimated drag coefficient is *not* physically meaningful over a 6-hour arc: it runs to 5.26 for S3A and even *negative* (−3.20) for S3B, against a physical expectation of $C_D \approx 2.2$. The reason is observability: over only ~4 revolutions the integrated drag signal is small and strongly correlated with the along-track components of the initial state, so the normal-equation system is ill-conditioned in the C_D direction and the parameter absorbs unmodelled along-track error rather than true ballistic information. Reliable C_D (or ballistic-coefficient) estimation requires multi-day arcs, and/or estimating a piecewise/empirical along-track acceleration, and proper observation weighting — see §13.

12 Error budget, interpretation and physical accuracy

12.1 Why the along-track channel dominates

A small persistent acceleration error $\delta a_{\parallel}$ in the flight direction, plus any along-track velocity error $\delta v_{\parallel,0}$ in the seed, integrates into a position error that grows super-linearly:

$$\delta r_{\parallel}(t) \approx \tfrac{1}{2}\langle \delta a_{\parallel} \rangle t^2 + \delta v_{\parallel,0}\, t \tag{16}$$

The quadratic term is the signature of a secular acceleration error; the observed S3A growth (~18 m at 1 h → ~120 m at 12 h → ~310 m at 24 h) is close to this t^2 shape, which is itself strong evidence that the residual is a *dynamical* (drag) effect and not random noise. By contrast, the conservative perturbations produce *bounded, oscillatory* radial and cross-track errors, which is exactly what Figures 1–2 show.

12.2 Dominant error sources (physical)

Table 5. Contributions to the along-track prediction error, largest first.

Source	Nature	Approx. effect	Mitigation
Thermospheric density	NRLMSISE-00 climatology + constant (not true daily) F10.7/Ap	dominant	Real historical space weather; density calibration
Ballistic coefficient C_DA/m	Fixed cannonball; true A varies with attitude	large	Box-wing/attitude model; estimate over long arc
Seed state error	Single state vs fitted arc	moderate	Batch fit (removes ~8×, §11)
Geopotential truncation (30×30)	Conservative, mostly radial/cross	small	Raise to 70×70
Solar/lunar ephemeris, SRP C_R	Conservative, periodic	small	JPL DE ephemeris; estimate C_R
Omitted EOP polar motion	Frame, ~metre-level	small	IERS finals2000A EOP file

12.3 Accuracy in context

For a propagation seeded from a *single* state with *no* fit and *climatological* space weather, a 24-hour total error of ~0.15–0.17 km RMS at 800 km is a strong result: it corresponds to roughly 2 m of growth per ~100-minute revolution, i.e. the dynamics reproduce each orbit to about one part in 3×10^6 of the orbit circumference. The

POEORB reference itself (~cm) is negligible in this budget, so the numbers above are the engine's own error, not the reference's. With a fitted arc the instantaneous accuracy is at the few-metre level (Table 4).

13 Conclusions and recommended physics refinements

The validation against independent, GNSS-derived Sentinel-3 precise orbits demonstrates that the conservative dynamics (geopotential, luni-solar attraction, tides, relativity) and the GCRF ↔ ITRF frame chain are correct at the metre level, and that the residual prediction error is physically localised in the drag/ballistic channel. The engine is therefore a sound basis for both long-term trajectory prediction and orbit determination. The following refinements, in physical priority order, would most improve accuracy:

1. **True space-weather drivers.** Replace constant F10.7/Ap with the actual historical daily indices for the arc (CelesTrak/CSSI). This directly attacks the dominant density error and should substantially reduce the along-track growth.
2. **Ballistic/empirical estimation over long arcs.** Estimate a ballistic scale factor (or a piecewise-constant / empirical along-track acceleration) over multi-day arcs, where C_D becomes observable — resolving the ill-conditioning found in §11.
3. **Attitude-dependent area (box-wing).** Replace the cannonball cross-section with a panel model for both drag and SRP, since the true projected area varies along the orbit.
4. **Higher-fidelity conservative models.** Raise the geopotential to 70×70, adopt a JPL DE ephemeris for Sun/Moon, and add an IERS EOP file (polar motion + UT1 series) — each trims the few-metre radial/cross floor.
5. **Formal covariance.** Propagate the estimation covariance to attach realistic uncertainty bounds to predictions (essential for conjunction/decay-window work).
6. **Direct GNSS measurement fit.** Fit to the raw onboard GPS observations (the Sentinel-3 GNSSRD product) rather than the reduced POEORB positions, closing the loop with the GNSS data directly.
